```
FILE="addressbook.txt"

createAddressBook()
{
  if [ -e "$FILE"] ;
  then
       echo "Address book already exists."
  else
       touch "$FILE"
       echo "Address book created."
  fi
}

viewAddressBook()
{
  if [! -s "$FILE"];
  then
      echo "Address Book is empty or does not exists."
  else
     echo "------Address Book----------"
     cat  "$FILE"

  fi
}

insertRecord()
{
   echo "Enter record in format:Last Name,First Name,E-mail,Phone"
   read record
   echo "$record">>"$FILE"

   echo "Record Inserted."

}

deleteRecord()
{
 echo "Enter a search string(e.g name or e-mail) to find the record to delete:"
 read search
 grep -n "$search" "$FILE"

 echo "Enter line number to delete:"
 read line
 sed -i "${line}d" "$FILE"

 echo "REcord deleted!"
}
modifyRecord()
```

```
{
 echo "enter a search string(e.g name or e-mail) to find the record to modify:"
 read search
 grep -n "$search" "$FILE"

 echo "enter lne number to modify:"
 read line
 echo "enter new record (Last Name,First Name,E-mail,Phone):"
 read newREcord
 sed -i "${line}s/.*/$newRecord/" "$FILE"
 echo "REcord modified"
}

exitProgram()
{
echo "Exiting ......Good Bye....."
exit 0
}
while true; do
    echo "==========Address Book Menue"
    echo "1. Create Address Book"
    echo "2. View Address Book"
    echo "3. Insert A  Record"
    echo "4. Delete A REcord"
    echo "5. MOdify A Record"
    echo "6. Exit"
    echo "Enter your choice"
    read choice

    case $choice in
        1)createAddressBook ;;
        2)viewAddressBook ;;
        3)insertRecord ;;
        4)deleteRecord ;;
        5)modifyRecord ;;
        6)exitProgram ;;
        *) echo "Invalid choice.Please selet 1-6" ;;

   esac
done
```

---------output----------

==========Address Book Menue
1. Create Address Book
2. View Address Book
3. Insert A Record
4. Delete A REcord
5. MOdify A Record
6. Exit
Enter your choice
1
./addbook.sh: line 5: [: missing `]'
Address book created.
==========Address Book Menue
1. Create Address Book
2. View Address Book
3. Insert A Record
4. Delete A REcord
5. MOdify A Record
6. Exit
Enter your choice
3
Enter record in format:Last Name,First Name,E-mail,Phone
Jahagirdar Gauri gauri@gmail.com 9284464731
Record Inserted.
==========Address Book Menue
1. Create Address Book
2. View Address Book
3. Insert A Record
4. Delete A REcord
5. MOdify A Record
6. Exit
Enter your choice
2
./addbook.sh: line 16: [!: command not found
------Address Book----------
Jahagirdar Gauri gauri@gmail.com 9284464731
==========Address Book Menue
1. Create Address Book
2. View Address Book
3. Insert A Record
4. Delete A REcord
5. MOdify A Record
6. Exit